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Enhancing University Admissions through Scalable and Fair Machine Learning Models: A Case Study from Yemeni Universities

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OUTLINES

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Research Problem & Motivation

A critical **problem** in developing countries such as Yemen is that limited resources and an increase in the number of applicants worsen the variations in higher education admissions [10], [22].

Inefficient traditional evaluation metrics

Relying heavily on traditional evaluation metrics suffers from *fairness* and *efficiency* issues. This leads to delays and a high potential for human error and subjective biases.

Bias in Admissions

Current systems heavily rely on metrics such as GPA and standardized test scores, which *may inadvertently disadvantage students from underserved regions* or specific socioeconomic backgrounds, perpetuating existing inequalities.

Demand for Fair Solutions

Therefore, we need a scalable and data-driven admissions framework to solve this problem. It should improve transparency, support fair decisions, and handle large numbers of applicants efficiently [4], [7], [14], [23]– [25].

Research objectives

This paper aims:

1 **Develop Accurate ML Models**

To **apply ML ensemble models for optimizing the admission process** by developing a *scalable* and *fair* framework of university admissions in Yemeni universities.

2 **Integrate Fairness Metrics**

To embed and optimize for *fairness* metrics, specifically *Disparate Impact Ratio* (DIR) and *Equal Opportunity Difference* (EOD), **ensuring unbiased admissions across diverse demographic groups**.

3 **Ensure Scalability**

To design and test models that are computationally efficient and suitable for deployment in low-resource computing environments common in developing regions, without compromising performance.

Methodology

1

Data Preprocessing

Rigorous preprocessing involved normalization to scale features, one-hot encoding for categorical variables, and robust outlier removal using statistical methods to enhance model stability and performance.

2

Model Selection

We explored a range of established machine learning algorithms: Logistic Regression (LR), Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), and Gradient Boosting (GB).

3

Fairness-Aware Optimization

To mitigate bias, we integrated fairness objectives by employing various penalized loss functions during model training, specifically designed to reduce disparate impact on protected attributes.

Evaluation Metrics

Performance

Evaluated using standard classification metrics: *Accuracy* (overall correctness), *Precision* (minimizing false positives), Recall (minimizing false negatives), *F1-Score* (balance of precision and recall), and *AUC-ROC* (model's discriminative power).

Fairness

Quantified using *Disparate Impact Ratio (DIR),* measuring proportional selection rates across groups, and *Equal Opportunity Difference (EOD),* assessing equality in true positive rates for different sensitive attributes.

Scalability

Assessed by analyzing *model runtime* (computational efficiency) and *memory usage during training* and inference, crucial for deployment in resource-constrained environments.

Evaluation Metrics

Table I Evaluation Metrics and Descriptions

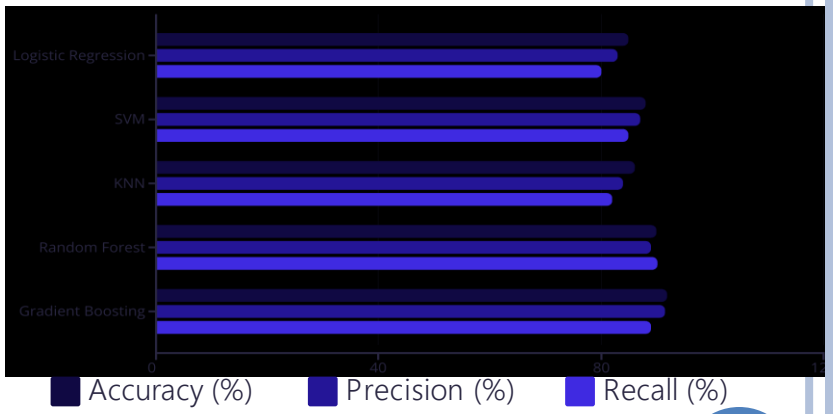
Metric	Description
Accuracy	Proportion of correct predictions over the total number of predictions.
Precision and Recall	Precision measures the accuracy of positive predictions, while recall measures the proportion of true positives correctly identified.
F1-Score	Harmonic mean of precision and recall, providing a balanced performance measure.
AUC-ROC	Evaluates the trade-off between true positive and false positive rates, with higher values indicating better classification capability.
Bias and Fairness Metrics	• **Disparate Impact Ratio (DIR)**: Let's face it - acceptance rates shouldn't depend on demographics. That's why we track DIR, which essentially answers: "Are some groups getting approved more often than others?" It's our first reality check for fairness. • **Equal Opportunity Difference (EOD)**: Imagine two equally qualified students from different backgrounds. EOD tells us if our system spots both candidates equally well. Specifically for admissions, we measure recall gaps - because missing qualified applicants hurts everyone.

RESULTS: PERFORMANCE

Table IV performance metrics of ML models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)
Logistic Regression	78.3	80.1	76.5	78.3	79.0
Random Forest	89.5	88.9	90.2	89.6	91.0
Gradient Boosting	92	91.5	90.8	91.2	93.0
Support Vector Machine	85.0	84.5	85.7	85.1	86.5
K-Nearest Neighbors	82.1	81.2	83.0	82.1	83.5

- Gradient Boosting: 92% accuracy, 91.5% precision
- Random Forest: strong recall (90.2%)
- Ensemble models outperform traditional methods



RESULTS: FAIRNESS

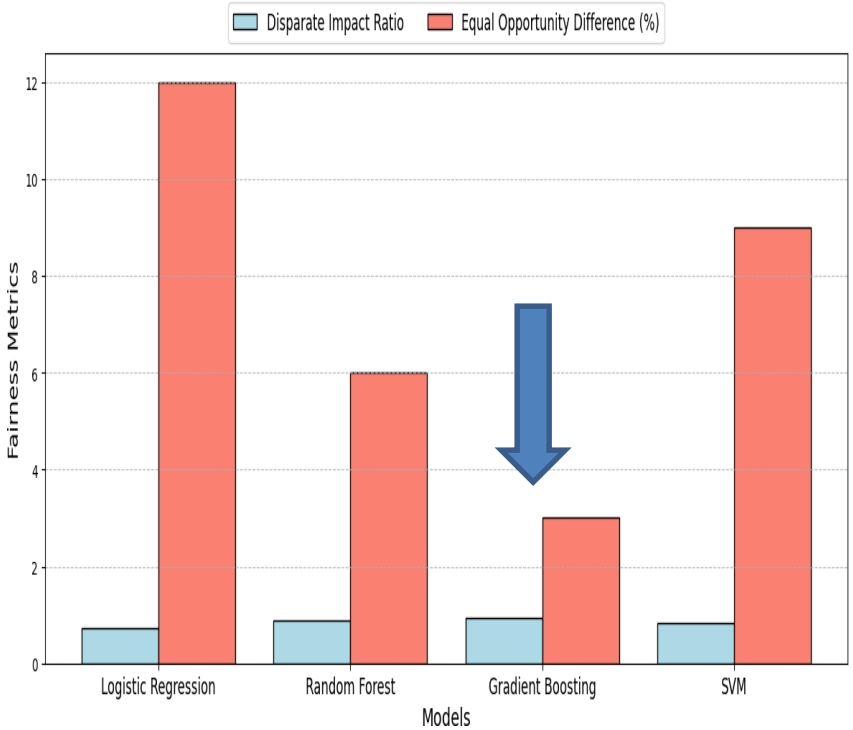


Fig. 1. Bias Reduction Metrics Across Models

- Gradient Boosting reduced bias by 20%
  DIR: 0.94 and EOD: 3%
- Logistic Regression had lower fairness scores

Results: Scalability

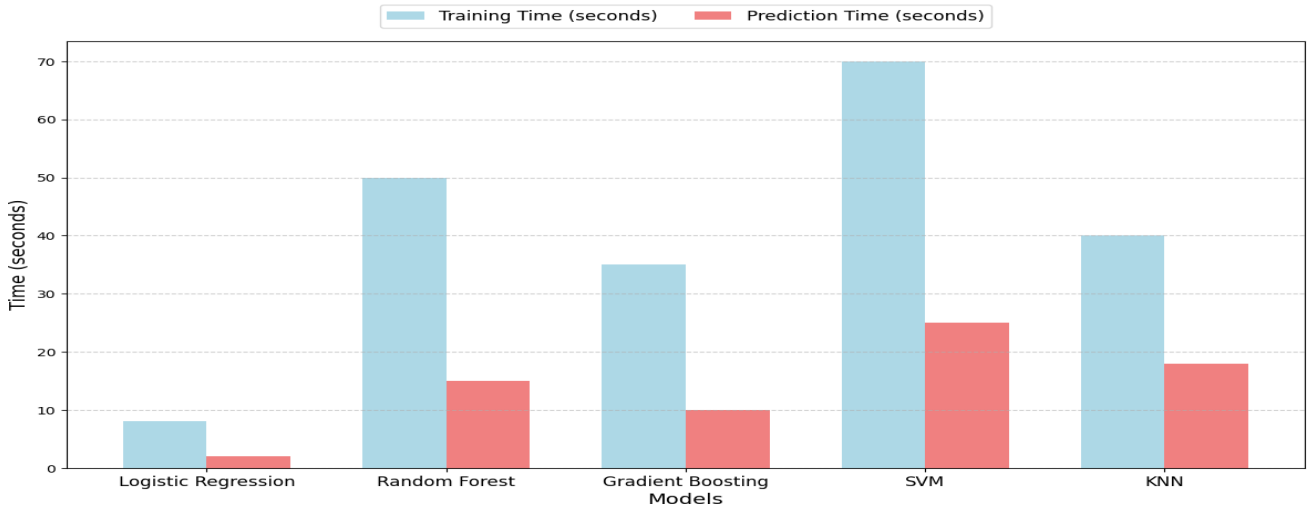


Fig. 2. Scalability Metrics for ML Models

- GB and RF: efficient runtimes under 50s

50s

GB Runtime

Gradient Boosting models demonstrated efficient training and inference times, typically completing within 50 seconds. This makes them highly practical for iterative development and deployment.

50s

RF Runtime

Similarly, Random Forest models exhibited comparable runtime efficiency, ensuring rapid processing of admission applications, even with larger datasets or frequent retraining cycles.

RESULTS: SCALABILITY

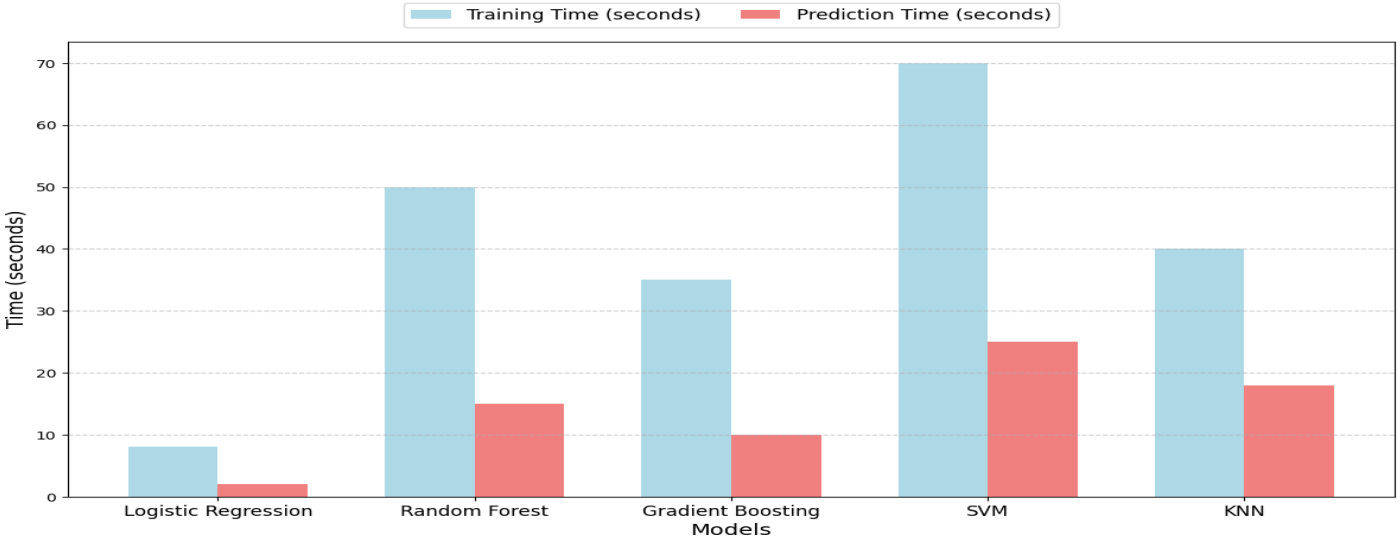


Fig. 2. Scalability Metrics for ML Models

- Suitable for <2GB RAM environments-
- SVM and KNN less scalable

2GB

Less Scalable

Memory Footprint

Both Gradient Boosting and Random Forest models proved to be suitable for environments with limited computational resources, operating effectively with less than 2GB of RAM during execution.

SVM & KNN

Conversely, SVM and KNN models showed higher computational demands, making them less suitable for deployment in low-resource settings prevalent in many Yemeni universities.

Feature Importance

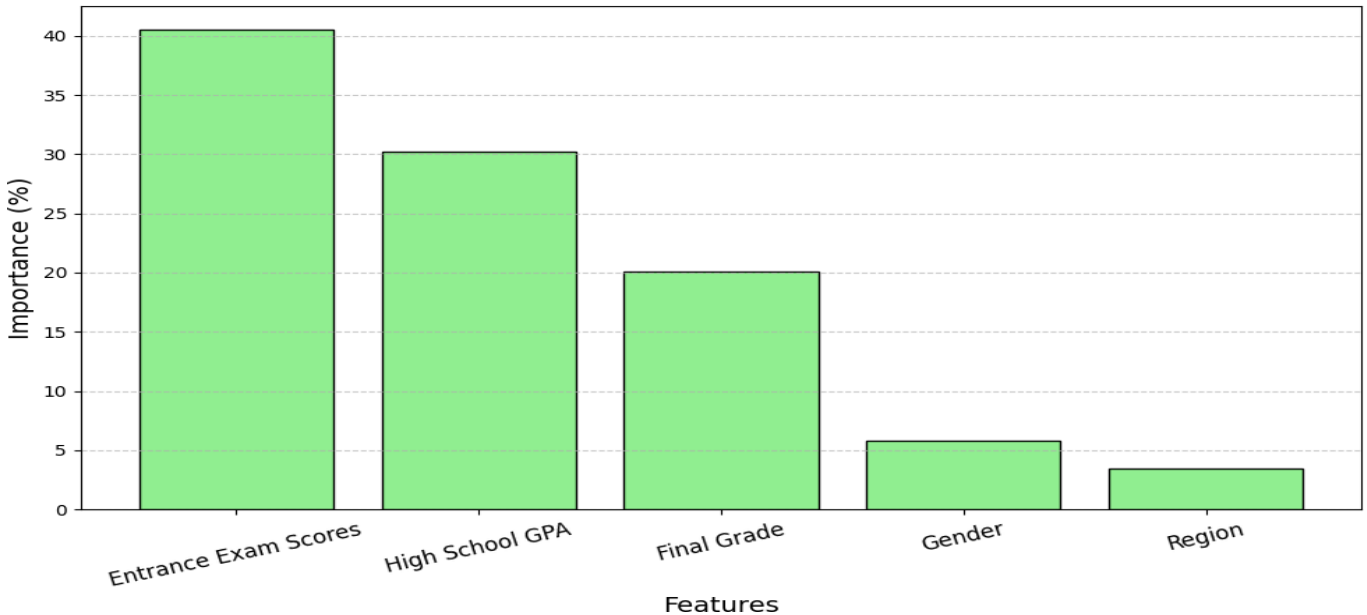


Fig. 3. Feature Importance from Random Forest

- Entrance exam scores and GPA: top predictors (70% influence)
- Geographic and parental education also relevant

Discussion

- Ensemble models are fair, interpretable, and scalable
- Effective even with low computational resources

CONTRIBUTIONS

- First fairness-aware ML study in Yemen's context
- Integrates Arabic-language and conflict-sensitive processing

Future Work

- Include extracurricular, leadership, location data
- Explore hybrid and real-time adaptive systems

Thanks for Your Attention

Fair admissions are not just a goal—they're a responsibility.

Q&A